```
import java.util.Arrays;
import java.util.Random;

public class Comparison {

	public static void main(String[] args) {

		int size = 1000;
		int[] original = new int[size];

		Random random = new Random();

		// Generate one random dataset
		for (int i = 0; i < size; i++) {
			original[i] = random.nextInt(10000);
		}

		// Give each algorithm the same data
		int[] selectionArray = Arrays.copyOf(original, size);
		int[] insertionArray = Arrays.copyOf(original, size);
		int[] bubbleArray = Arrays.copyOf(original, size);
		int[] mergeArray = Arrays.copyOf(original, size);

		// Selection Sort
		long start = System.nanoTime();
		SelectionSort.sort(selectionArray);
		long selectionTime = System.nanoTime() - start;

		// Insertion Sort
		start = System.nanoTime();
		InsertionSort.sort(insertionArray);
		long insertionTime = System.nanoTime() - start;

		// Bubble Sort
		start = System.nanoTime();
		BubbleSort.sort(bubbleArray);
		long bubbleTime = System.nanoTime() - start;

		// Merge Sort
		start = System.nanoTime();
		MergeSort.mergeSort(mergeArray, 0, mergeArray.length - 1);
		long mergeTime = System.nanoTime() - start;

		// Display results
		System.out.println("===== SORTING COMPARISON =====");
		System.out.println("Array Size: " + size);
		System.out.println();

		printResult("Selection Sort", selectionTime, selectionArray);
		printResult("Insertion Sort", insertionTime, insertionArray);
		printResult("Bubble Sort", bubbleTime, bubbleArray);
		printResult("Merge Sort", mergeTime, mergeArray);
	}

	static void printResult(String name, long time, int[] arr) {
		System.out.println(name);
		System.out.println("Execution Time: " + time + " ns");
		System.out.println("Execution Time: "
				+ time / 1_000_000.0 + " ms");
```

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```
            System.out.println("Sorted Correctly: " + isSorted(arr));
            System.out.println("----------------------------");
        }

        static boolean isSorted(int[] arr) {
            for (int i = 0; i < arr.length - 1; i++) {
                if (arr[i] > arr[i + 1]) {
                    return false;
                }
            }
            return true;
        }
}
```

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